



On-Prem Cloud Platform Functional & Technical Reference

An accessible reference for all stakeholders to fully grasp the solution's scope and impact.

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1. Introduction

This document provides technical and functional specification of the **On-Prem Cloud Platform (OPCP)**, a modular and sovereign cloud infrastructure solution designed by OVHcloud. It describes the architecture, features, and operational models required to deploy and manage a full-stack cloud environment directly on customer premises.

The objective of this specification is to provide a reference for all stakeholders involved in the design, deployment, operation, and consumption of OPCP including technical teams, decision-makers, partners, and compliance officers. It clarifies the scope of the solution, outlines supported use cases, and defines the components, interfaces, and services that make up the platform.

This document does **not cover internal engineering implementation details**, but rather focuses on describing the **capabilities, integration models, and expected behaviors** of the solution, in line with OVHcloud's standards and best practices. Where relevant, it points to dedicated document for in-depth topics such as support, professional services, or onboarding methodology.

The scope includes:

- Infrastructure footprint and scalability models (Nano, Standard, Large)
- Software architecture and orchestration layers
- Security, compliance, and interoperability requirements
- User roles and operational responsibilities
- Deployment and integration patterns, including hybrid and edge

By consolidating this information, this document ensures consistency across deployments and facilitates alignment between OVHcloud teams and customer organizations during all stages of the OPCP lifecycle from initial assessment to full production delivery.

2. Solution Overview

a. About On-Prem Cloud Platform (OPCP)

OPCP is the result of over 25 years of expertise in automating and industrializing Cloud Services. It's a turnkey hardware and software solution that delivers advanced cloud capabilities directly from your own premises.

Designed with modern sovereignty requirements and cloud public attributes in mind, it lets you stay in control while keeping pace with the latest cloud innovations.

Built on open-source foundations and equipped with a local control plane, it offers extensive automation and tooling to maximize independence and interoperability. A rich software catalog aligned with OVHcloud's mainstream services—provides access to advanced, ready-to-deploy technologies.

Its modern, intuitive interface empowers end-users to design secure, scalable, robust, and efficient environments for all types of applications.

The solution is built to flex from ultra-compact setups for space-tight Edge environments to high-power, high-density configurations that can handle the heaviest workloads, including GPU-hungry AI applications. Connected or disconnected.

Finally, OPCP is deeply compatible by design with OVHcloud's ecosystem and deployment models, offering the broadest possible range of hybrid and cloud-native capabilities.





b. Primary Use Cases

OPCP is built to adapt whether you're delivering services, deploying at the edge, running AI workloads, or operating in fully disconnected environments. Its flexibility and depth of integration make it the ideal foundation for a wide range of scenarios, combining cloud agility with full control, performance, and sovereignty.

Cloud Services Delivery

Provide ready-to-use Cloud Services directly from your own premises just like a public cloud, but fully under your control. Ideal for both internal teams and external customers.

Edge Computing

Bring the cloud wherever it's needed even in the most remote locations. Compact, autonomous deployments ensure continuity even when connectivity is lost.

Hybrid Cloud

Extend your public cloud environment into your on-prem datacenter. Enjoy the best of both worlds including seamless, secure integration with sensitive SNC environments.

AI Workloads

Run AI at scale with native support for GPU acceleration and machine learning frameworks while keeping training and inference data securely on-site.

Air Gap

Operate a fully self-contained, secure cloud with an embedded control plane. No external connectivity required, the control plane is onboard.

Appliance

Deploy OPCP as a ready-to-use appliance. In partnership with independent software vendors (ISVs), OPCP can be bundled as a turnkey, end-to-end solution combining infrastructure, automation, and a specific application stack into a single, integrated package.



c. Audiences

OPCP empowers every stakeholder in your on-prem cloud journey from Solution architect, datacenter operators and system administrators to Platform engineers, DevOps and SREs and all supporting functions involved. Each profile benefits from intuitive tools, built-in automation, and secure-by-design features tailored to their activities.

Whether you're managing infrastructure, delivering services, or consuming them, OPCP ensures a seamless, efficient, and future-ready experience.

The solution can be fully operated autonomously by your teams, or partially or entirely managed by OVHcloud experts or qualified partners adapting to your preferred operating model.

Moreover, five main roles are implemented in OPCP control plane:

DC Operator

Responsible for the physical layer, the DC Operator ensures hardware components are functioning properly installing servers, replacing disks, and monitoring power, temperature, and connectivity.

SysOps

The SysOps oversees the undercloud: enrolling servers, setting up networks and projects and allocating resources. They also manage firmware upgrades, control plane updates, and monitor capacity and availability.

Cloud Provider

The Cloud Provider publishes and manages Cloud Services from the software catalog. They create Cloud users accounts, assign rights and quotas, monitor service health, and ensure compliance with integrated security, audit, and observability standards. They also manage usage tracking for cost control and chargeback.

Service packager

The Service Packager builds ready-to-deploy service bundles, integrating binaries, containers, and Terraform code into the software catalog (OPCP Cloud Store). By aligning with OPCP standards, they guarantee that services are secure, observable, and fully manageable.

Cloud Users

Cloud Users access resources via a unified self-service portal or API provisioning VMs, bare metal, AI services, databases, or complex multi-tier architectures. With a single account, they manage services, delegate quotas, fine-tune access control, and monitor usage. They can interconnect services securely, manage secrets, and build scalable environments autonomously.

Example responsibilities for teams:

DC Operator	SysOps	Cloud Provider
• Install and connect physical equipment (rack, cabling, power, network) • Monitor hardware health and perform maintenance • Enroll nodes into the platform • Apply procedures for hardware recycling or replacement	• Manage platform capacity and resource availability • Manage Network and project isolation • Allocate physical resources via usage quotas and role-based access control (RBAC) • Deploy native OS on baremetal nodes	• Manage tenant and account • Allocate and monitor services • Track usage, billing, and compliance

Service Packager	Cloud Users
• Design and package new Cloud Services • Define metadata, dependencies, and deployment scripts • Test and validate services before publishing • Manage service versions and related documentation	• Access services via UI or API • Create, configure, and operate cloud resources (VMs, storage, containers, etc.) • Monitor usage and alerts • Manage credentials, secrets, and network settings

3. Major technical features

OPCP delivers a full-stack cloud solution through three integrated technical layers, each playing a distinct role in automating, operating, and scaling on-premises cloud environments.

- **Layer 1: OPCP CORE – Plug & Cloud**

The foundational layer, providing the hardware, infrastructure tools, and automation PLCs required to deploy and operate a reliable, secure, and efficient cloud base.

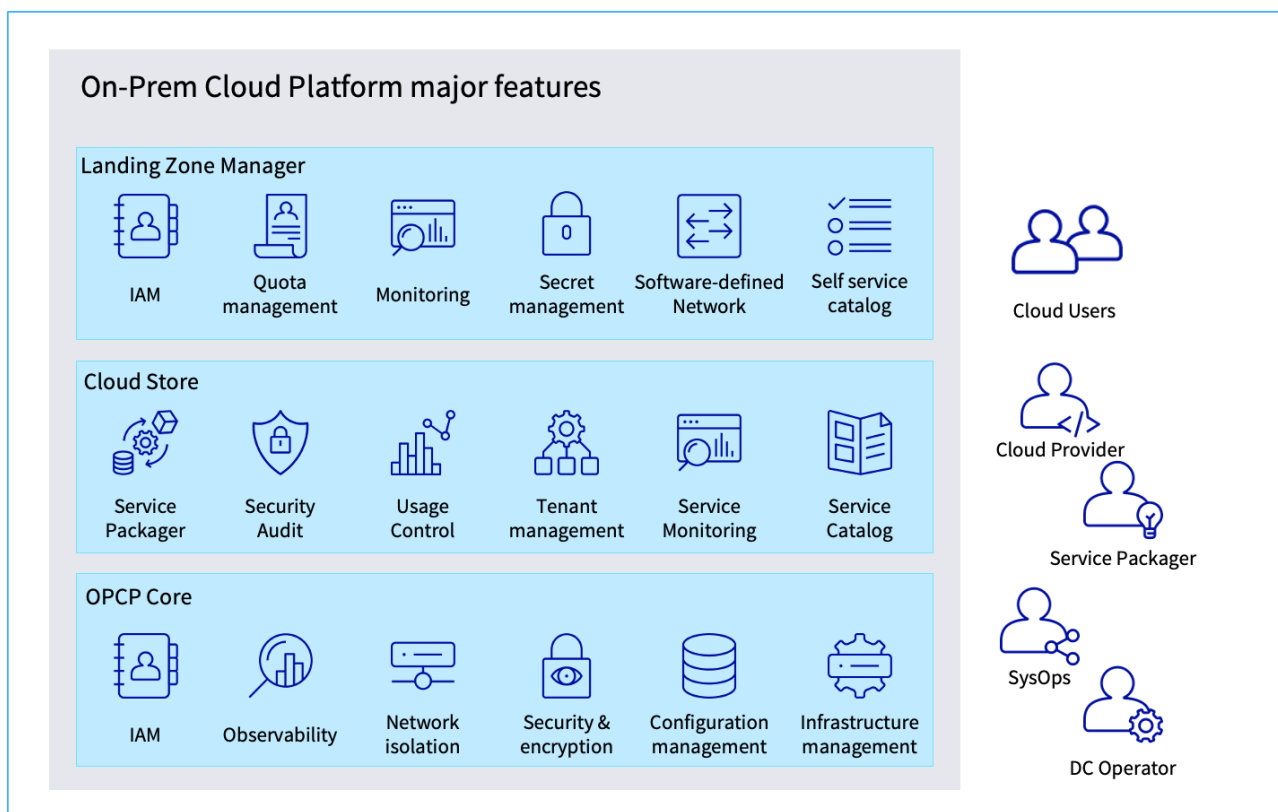
- **Layer 2: Cloud Store – Click and Deploy**

A comprehensive service catalog offering off-the-shelf software components and lifecycle management tools to provision, scale, and maintain Cloud Services.

- **Layer 3: Landing Zone Manager – Cloud & Play**

The user-facing orchestration layer, delivering an intuitive interface and API to deploy and manage application environments, enabling self-service, governance, and automation.

Together, these layers provide a robust, modular, and future-proof platform to support a wide range of cloud use cases from infrastructure management to application deployment and service consumption.



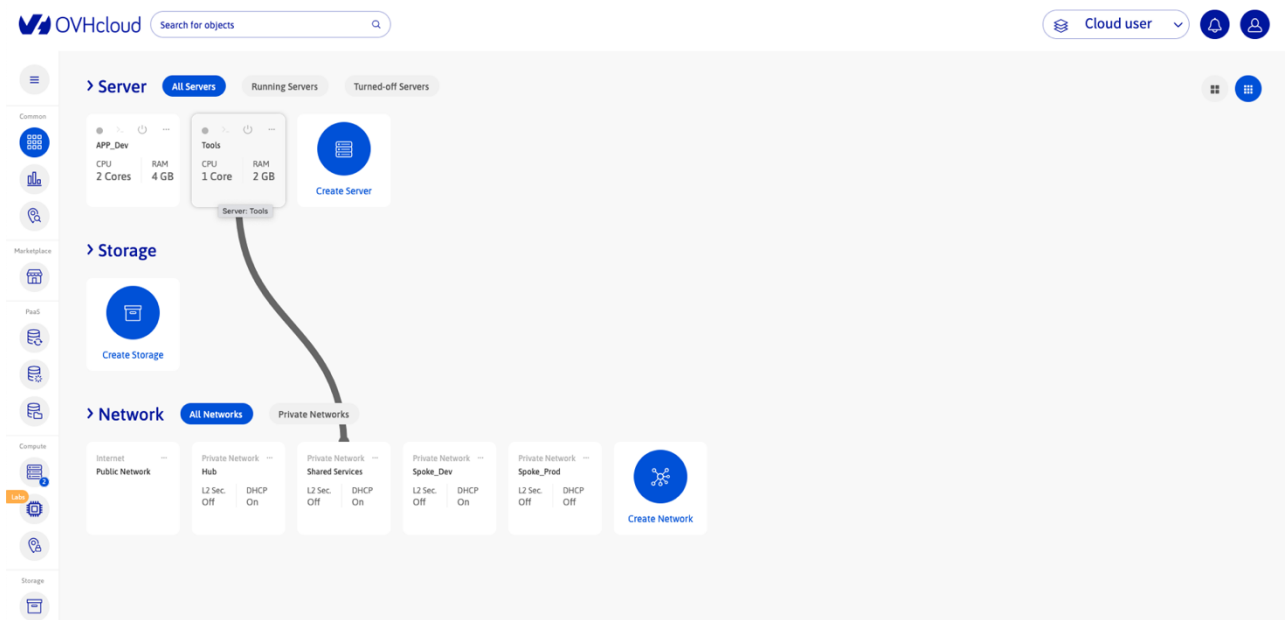


a. Landing Zone Manager

i. Overview

The Landing Zone Manager is the user-facing orchestration and governance layer of OPCP. It offers a unified, self-service experience to design, deploy, and operate cloud-native application environments on-premises securely, efficiently, and at scale.

Built for modern DevOps and platform teams, it abstracts infrastructure complexity while enforcing governance, allowing users to focus on delivering value. Whether via GUI or API, it provides a seamless interface to consume cloud resources in a consistent and controlled manner.



ii. Self-service portal & API

At the heart of the Landing Zone Manager is a powerful self-service portal and API. Users can access a curated catalog of cloud resources, provisioned by cloud store, and deploy them with just a few clicks or lines of code.

Templates, versioning, and parameterization enable rapid, standardized deployments, while native integration with identity, quota, and security controls ensures safe and compliant operations.

iii. Tools

To support enterprise-grade governance and automation, the Landing Zone Manager natively integrates with the following key services:

Identity and Access Management (IAM)

Role-based access control (RBAC), single sign-on (SSO) and Multi-Factor Authentication (MFA) ensure users can only access what they're entitled to, aligned with organizational policies.

Quota Management

Resource consumption can be limited and monitored per user, allowing fine-grained cost control and service isolation.

Software-Defined Networking

Native SDN integration provides full control over network segmentation, isolation, and interconnectivity:

- **Private network isolation** to ensure traffic separation.
- **Dynamic IP management and virtual routing** to support scalable architectures.
- **Integrated firewalling policies** to control ingress, egress, and internal flows.

Support for hybrid connectivity and secure service exposure when needed.

Monitoring

Integrated observability tools provide full insight into platform activity and service performance, including:

- **Usage:** Real-time and historical metrics on resource allocation.
- **Consumption:** Fine-grained usage reporting per workload or tenant.
- **Logs:** Centralized logging for auditing, debugging, and compliance.

Secret management

Secure, centralized storage and lifecycle management of secrets such as API keys, certificates, and passwords available through native interfaces and automation-friendly APIs.



b. Cloud Store

i. Overview

The Cloud Store is the central component for managing and delivering Cloud Services in OPCP. It provides a CSP-like experience to publish, consume, and operate services from core infrastructure to advanced workloads like Kubernetes, AI, and databases.

It enables internal teams and service operators to offer catalog-based Services across apps with built-in lifecycle management, security, observability, and governance turning the On-Prem Cloud Platform into a fully-fledged, multi-tenant cloud back-end.

ii. Service catalog

The catalog is modular, extensible, and governed. It supports multiple service types and publication scopes:

- **Open-source:** Pre-integrated and validated services based on open-source software.
- **License:** Pre-integrated and validated services based commercial software requiring a license for use.
- **Closed source:** Managed services that are necessarily operated by a software vendor, an authorized partner or OVHCloud.
- **Internal:** Service packaged by organization itself for internal use.

Feature Stages – Services can be labeled alpha, beta, GA, or deprecated, allowing progressive rollouts and lifecycle control.

iii. Service management features

Service can benefit from lifecycle management features via UI or API:

- **Deploy / Remove** – Launch or decommission services using templated configurations.
- **Scale / Downscale** – Adapt to demand by resizing allocated infrastructure.
- **Upgrade Software** – Apply safe, versioned software upgrades with rollback support.
- **Backup / Restore** – Manage persistent data and configuration snapshots.
- **Tenant Management** – Delegate granular access control for each service.
- **Quota Management** – Enforce resource limits at app level

Note: Service can benefit from different management features depending on their type, maturity, and overall packaging. Each application can leverage the features most relevant to its context."

iv. Usage control

Track, report, and limit resource usage per service or app enabling chargeback, show back, and quota enforcement.

v. *Security & Audit features*

Security is enforced across services via integrated mechanisms:

- **KMS Integration** – Native support for key encryption and management.
- **HSM Support** – Optional integration with hardware security modules.
- **Logs** – Centralized logging for service actions and access.
- **Inventory** – Real-time view of active services, configurations, and interdependencies.

vi. *Service monitoring*

Service-level observability is built-in, using standard tools (e.g., Prometheus, Grafana) for metrics, health status, and alerts ensuring transparency and operational efficiency.

vii. *Service Packager*

The Cloud Store includes a built-in **Service Packager** feature that enables users (with appropriate roles) to onboard and publish new services into the catalog. By packaging binaries, containers, or Terraform modules, and applying OPCP-compliant metadata (security, observability, lifecycle hooks), Service Packagers can expose reusable, manageable, and secure services within the platform. This allows organizations to expand their cloud offering, internally or for resale, while maintaining governance and interoperability across the stack.

viii. *Service availability*

OVHcloud provides within the Cloud Store a rich and evolving portfolio of ready-to-deploy services, carefully curated to meet the diverse needs of cloud environments.

This software catalog is built on a foundation of trusted open-source technologies, complemented by solutions from strategic technology partners, as well as a selection of OVHcloud's own mainstream public services.

The goal is to deliver a robust, sovereign, and interoperable software layer that enables a wide range of use cases, from container orchestration and object storage to managed databases and AI platforms.

The catalog includes a variety of pre-integrated solutions, from Independent Software Vendor, such as cyber defense platforms, Web Application Firewalls (WAF), data analytics stacks, and Large Language Models (LLMs).

It also offers the flexibility to deploy industry-specific business applications tailored to vertical needs, helping organizations accelerate their digital transformation while retaining full control over their infrastructure.

The catalog is accessible through the Cloud Store portal and is continuously updated with new services and features aligned with technological advancements and market demands.

[Learn more by exploring OPCP Cloud Store Datasheet.](#)



c. OPCP CORE

i. Overview

OPCP CORE is the foundational engine of the platform, the layer responsible for infrastructure orchestration, configuration management, security, and visibility. It centralizes infrastructure operations and provides intuitive dashboards and interfaces for day-to-day management and diagnostics.

Key interfaces and tools include:

- Dashboard – Access panel.
- NetBox – Network and datacenter modeling.
- Horizon – Web UI for resource allocation.
- Grafana – Dashboards for metrics, observability, and alerting.
- Keycloak – IAM.

This core layer can operate as a standalone foundation, enabling organizations to manage infrastructure independently: deploy networks, allocate bare metal resources, and operate them at scale in an industrial-grade manner.

ii. Infrastructure Management System

OPCP CORE includes a fully automated infrastructure management system that handles the full hardware lifecycle and network orchestration.

- **Enroll:** Automatically detect, check, trust, update and register new hardware.
- **Deploy:** Provision and configure hardware: Firmware, Network and OS
- **Recycle:** Reclaim, reset, and redeploy nodes as needed.
- **Network Automation:** Automate switch configuration, VLAN propagation.

iii. Configuration Management

Configuration management is a cornerstone of reliable infrastructure operations, ensuring consistency, visibility, and control across all deployed resources via NetBox.

- **CMDB** (Configuration Management Database): Central source of truth for assets and configurations.
- **DCIM** (Datacenter Infrastructure Management): Inventory and environmental monitoring.
- **IPAM** (IP Address Management): Centralized IP planning and allocation.

iv. Identity and access management

Robust identity and access management is essential to securing a multi-tenant cloud environment and simplifying user experience.

- Keycloak provides SSO (Single Sign-On), identity federation, and RBAC (Role-Based Access Control) across all OPCP components.
- Ensures unified authentication and authorization across the stack.

v. Observability

OPCP CORE includes native monitoring with **Prometheus** and **Grafana**, offering **preconfigured dashboards, alerts, and reports** for instant visibility into infrastructure and service health.

The observability stack is **extensible and interoperable**, allowing integration with external supervision, ticketing, or incident management tools (e.g., via APIs, webhooks, or exporters).

The observability stack is unified across the entire solution. It is hosted within OPCP Core, while the other layers simply consume its capabilities.

vi. Network isolation

OPCP supports virtualized, segmented, and policy-enforced networks.

Enables multi-tenant isolation, custom traffic flows, and firewalling across project at infrastructure level.

vii. Server Allocation & configuration

Server resources are provisioned and managed through **Horizon Interface**, with automation and quota enforcement.

You create project and allocate resources, and—when needed—directly use bare metal servers for specific workloads requiring full control or maximum performance.

viii. API / Infrastructure as Code

All core functions are accessible via **standard APIs**, enabling full automation and Infrastructure-as-Code workflows.

Compatible with Terraform, Ansible, and other IaC tooling for DevOps pipelines.

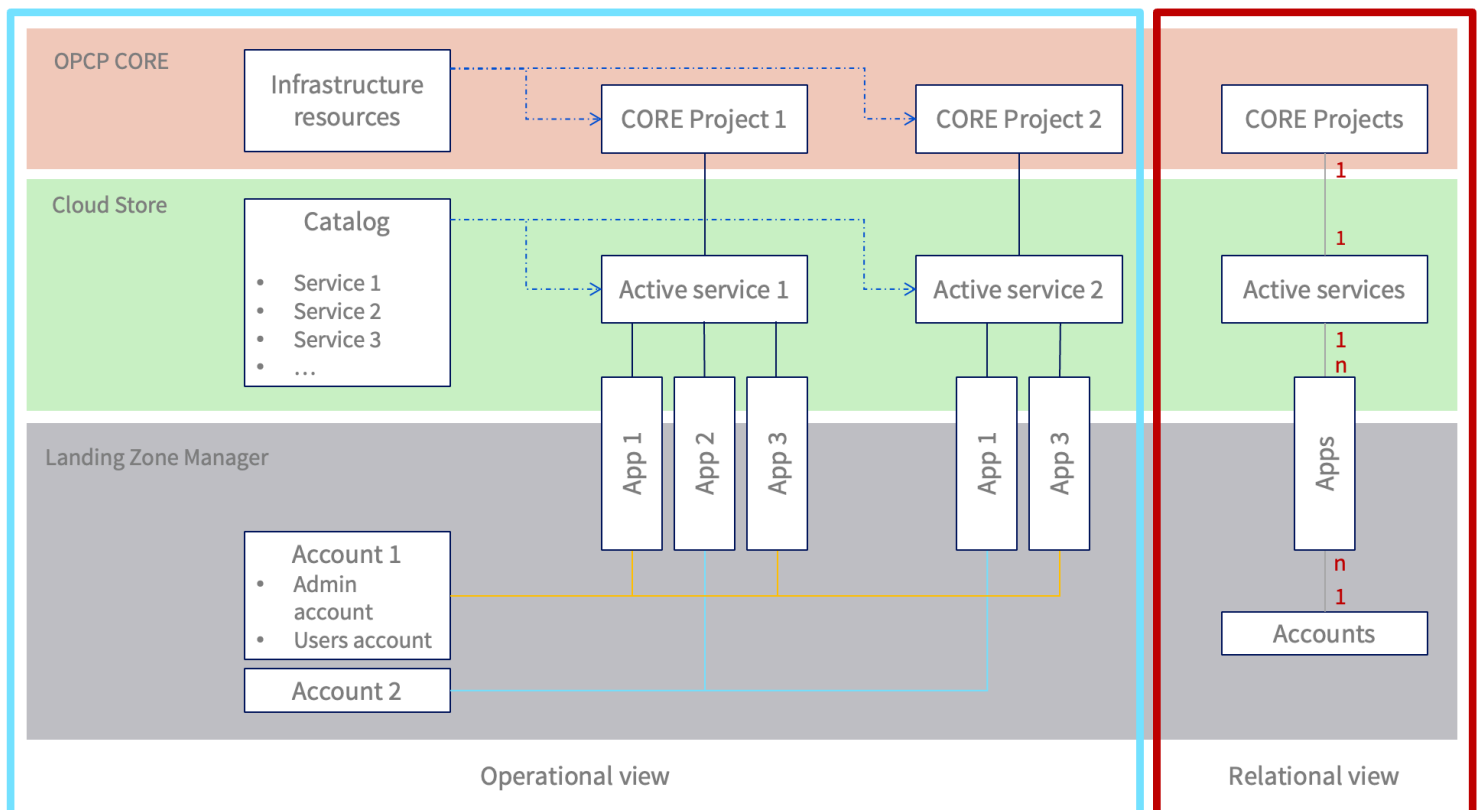
4. Resources hierarchy and access control

a. Resources hierarchy

OPCP defined a hierarchical model to organize and manage resources within an OPCP instance. For multi-instance management, see the *High Availability & Disaster Recovery* chapter.

This structure ensures proper ownership and policy access across resources. The main resource types in this hierarchy are:

- CORE Project
- Active service
- App
- Account



Instance view: the operational vision of resources as they are actually instantiated and consumed.

Relational view: the logical vision of the relationships and dependencies between each type of resource.

i. CORE Project

A CORE Project subdivides infrastructure resources into isolated pools consumed by Cloud store to host services.

Isolation between CORE Projects is guaranteed by network L3 VLAN thanks to a switch automatically managed by the system.

CORE Project is configured by SysOps role or automatically managed by Cloud Store to install Cloud Services.

ii. Active service

An active service is a service instantiated from a service available in the Cloud Store.

Multiple active services can be deployed from the same service. A service represents the operational instance of that service within a given CORE Project.

Its full lifecycle is managed through the Cloud Store and operated by the Cloud Provider role.

iii. App

Apps are subdivisions of an active service.

They provide logical isolation of independent environments within the same service.

Each app is dedicated to a single Account, ensuring strict separation and security.

iv. Account

An account is a group of users with rights and dedicated consumption space in the Landing Zone Manager.

Each account is logically and securely isolated from all others, ensuring complete separation of workloads.

Each account can consume active services in their apps regarding access granted and allocated quota by creating and managing workloads.

The Account is managed by the Cloud Provider role.

Access to workloads is enforced through fine-grained Role-Based Access Control (RBAC) policies. User permissions are scoped and securely managed within the Account repository, administered by the Cloud User Administrator role at the Account level.



b. Design access boundaries

Access boundaries in OPCP provide structural isolation across physical infrastructure, organizational accounts, and individual resources. Depending on operational models, regulatory constraints, and workload characteristics, each level of the resource hierarchy can be configured to support varying degrees of segregation and delegation.

i. Physical Resource Isolation

CORE Projects serve as physical isolation boundaries. They are typically aligned with sovereign infrastructure requirements, or dedicated hardware pools. In some deployments, each client or compliance domain may be mapped to a separate CORE Project to ensure full separation of compute and network paths at the physical layer. The level of granularity varies depending on security policies and hardware tenancy models.

ii. Organizational and Policy Domains

Accounts are used to logically partition resources by organization, client, or functional unit. Each account has its own IAM boundary, resource quotas, and visibility scope. Accounts may represent distinct business entities in multi-tenant environments, or separate departments in internal enterprise setups. In some cases, apps are also used to isolate workloads with different data residency or policy requirements, even if managed by the same team.

iii. Fine-Grained Access Control

At the Workload level, OPCP supports fine-grained IAM bindings. Resources such as virtual machines, containers, databases, and storage volumes can be independently managed and secured.

This level of granularity is particularly useful in shared environment scenarios, where multiple teams operate independently. However, access models at this layer tend to vary widely, from centralized administrative control to distributed, service-specific delegation models.



5. High Level Design

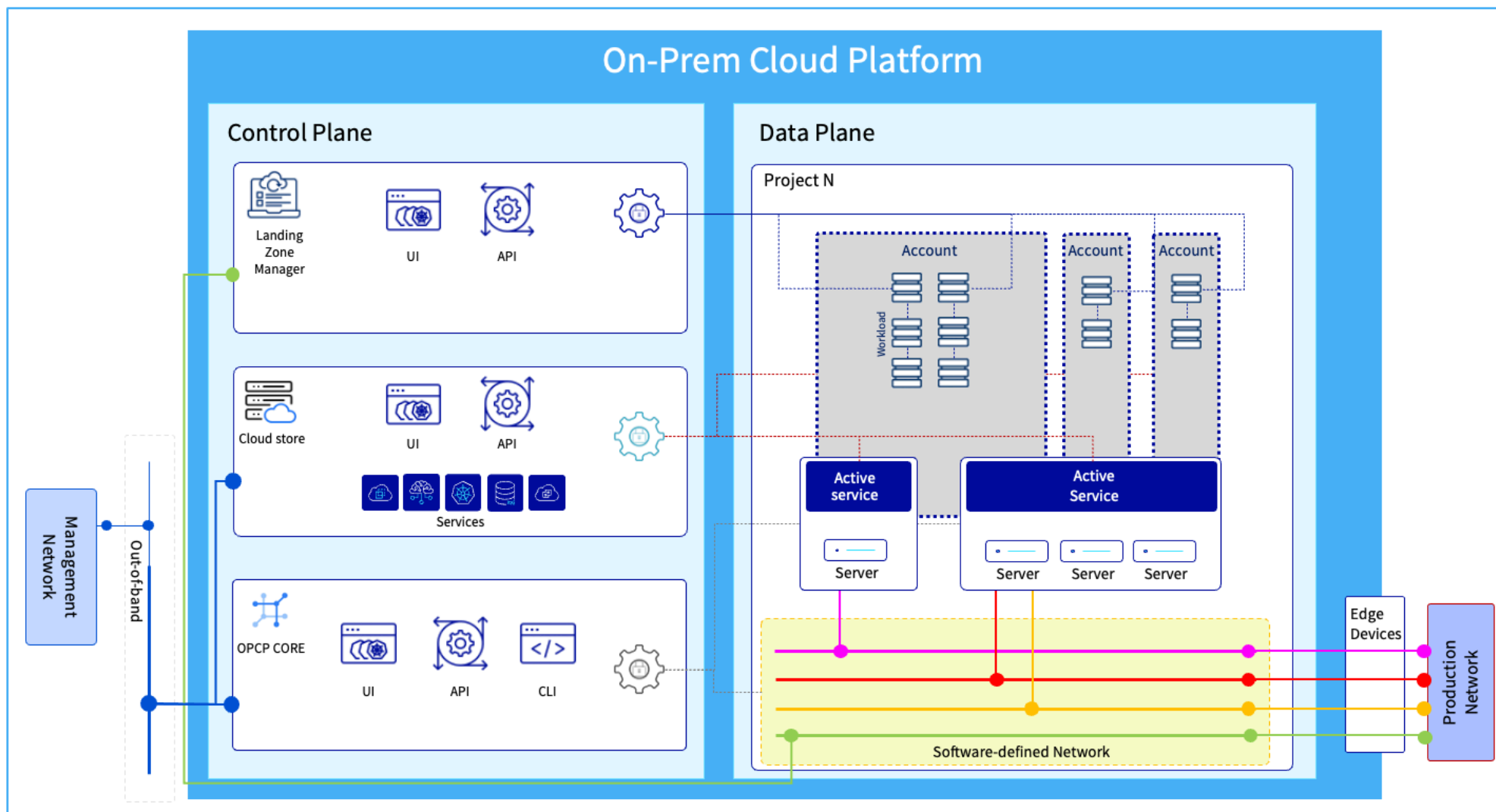
The OPCP reference design is modular and scalable by design, accommodating a variety of deployment models (nano, standard, large) and use cases, including edge computing, AI workloads, hybrid cloud, and cloud service provisioning.

It is structured around the following key components:

- **Base Infrastructure:** Pre-integrated racks with bare metal servers, redundant power supplies, and networking equipment.
- **Control Plane:** Core components responsible for deployment, orchestration, automation, and lifecycle management, including Landing Zone Manager, OPCP Core, and the Cloud Store.
- **Data Plane:** Customer workload resources, logically and physically isolated from control plane in app and project, ensuring secure operations and performance separation.



a. Conceptual architecture diagram





b. Physical architecture

i. Common physical components

Regardless of the deployment scale, OPCP relies on a consistent set of physical components. These elements form the foundation of the platform, providing the necessary functions for infrastructure

- **Controller:** The central “brain” of the infrastructure, managing, monitoring, and orchestrating cloud operations.
- **ToR IPMI** (Top-of-Rack IPMI switches): Secure, out-of-band connections for remote hardware management and troubleshooting without affecting production traffic.
- **ToR** (Top-of-Rack switches): The main data switches inside each rack, linking servers to each other and to the broader network.
- **Servers:** The computing and storage units that run applications, process data, and deliver services.
- **Edge devices:** The connection point between the on-premises cloud and external networks or the public cloud.
- **Spine switches:** The high-speed backbone interconnecting all racks, ensuring fast and reliable communication across the entire platform.

ii. Nano model Deployment

The Nano scale configuration is designed for minimal footprint deployments, ideal for constrained environments such as edge locations, branch offices, or sovereign installations. It offers a fully self-contained rack with all core OPCP components running within a single enclosure.

Smallest design possible:

- Controllers: 1
- ToR IPMI switch: 1
- ToR switch: 2
- Servers: N X Servers with $N = f(\text{rack_size})$
- Edge device: 0
- Spine switch: 0

Network capacity:

- 4x 25G per server
- 2x 10G local connection per rack

Notes:

100G local connection is possible by reconfiguring a server interface by using dedicated 100G (reserved UPLINK)

iii. Deployment Standard model

The Standard scale targets mid-size deployments, providing a balanced architecture that supports scalability, fault tolerance, and service diversity. With three dedicated racks, it ensures redundancy of core components while maintaining operational simplicity.

Intermediate design:

- Controllers: 1 per rack (x3)
- ToR IPMI switch: 1 per rack (x3)
- ToR switch: 2 per rack (x3)
- Servers: N X Servers with $N = f(\text{rack_size})$ (x3)
- Edge device: 2
- Spine switch: 0

Network capacity:

- 4x 25G per server
- 6x 100G total with EDGE: from 4x 100G to 6x 100G rack<->edge and any external connections from 10G to 100G

iv. Deployment large model

The Large-scale configuration is built for enterprise-grade or service provider environments, where high performance, capacity, and redundancy are critical. It enables horizontal expansion across multiple racks from 3 to 150+, supporting large volumes, extensive service catalogs, and advanced high availability and disaster recovery scenarios.

Large design:

- Controllers: 3
- ToR IPMI switch: 1 per rack
- ToR switch: 2 per rack
- Servers: N X Servers with $N = f(\text{rack_size})$
- Edge device: 2
- Spine switch: to define regarding number of racks. Mega spine can add for very large configuration

Network capacity:

- 4x 25G per server
- 6x 100G total with EDGE: from 4x 100G to 6x 100G rack<->edge and any external connections from 10G to 100G



c. Network integration

The On-Prem Cloud Platform is designed to seamlessly integrate with the customer's existing network infrastructure, ensuring secure, high-performance, and flexible connectivity across hybrid and multi-site environments. This integration is structured around the following key capabilities.

i. Seamless Integration

- OPCP is compatible with existing Layer 2 and Layer 3 network architectures.
- Native support for enterprise protocols such as BGP and VLAN.
- The platform can connect directly to the customer's core or distribution layer via standard uplinks (1G/10G/25G/100G) using customer-defined IP plans.

ii. Customizable Network Topologies

- Flexible design to adapt to existing segmentation policies (management, production, storage, DMZ).
- Supports both flat and segmented topologies, enabling integration into environments with strict network zoning and compliance requirements.
- Possibility to reuse existing subnets, route domains, or implement dedicated networks if required.

iii. Interoperability

- Compatible with existing firewalls, NIDS/NIPS, and proxy systems.
- Can be deployed behind existing perimeter security or within zero-trust architectures.
- Facilitates transparent traffic inspection and logging without duplicating security components.

iv. Automated Network Configuration and Discovery

- Network automation module automates the discovery and provisioning of network components during installation.
- Integration with CMDB/DCIM OPCP systems ensures alignment between logical and physical configurations.
- Reduces manual intervention during deployment and operations.

This approach allows customers to deploy OPCP without redesigning their network, while leveraging their existing investments in infrastructure, security, and governance.



d. High availability & Disaster recovery

High Availability (HA), resiliency, and Disaster Recovery (DR) are managed differently depending on the layer of the OPCP stack.

This layered approach to high availability and disaster recovery ensures that OPCP can meet the operational continuity requirements of critical workloads, whether deployed in isolated on-premises racks or extended across geographically separated data centers.

i. OPCP Core

For OPCP Core, high availability and resiliency are ensured at the solution level. Core components such as controllers, system services, and monitoring functions are deployed in redundant mode within a single site, with automatic failover in case of local hardware failure.

Network resiliency is provided through redundant switching and VLAN isolation. Disaster recovery at the Core layer relies on configuration and system state backups, which can be restored using local recovery mechanisms.

ii. Cloud Store

The Cloud Store is deployed with the same redundancy and backup mechanisms as the Core layer, including the Service catalog, service metadata, and app configurations. Cloud Services benefit from backup and restore features that allow the management of persistent data and configuration snapshots.

iii. Landing Zone Manager

The Landing Zone Manager system is deployed with the same redundancy and backup mechanisms as the Core layer, covering accounts and workload configuration data.

Workload high availability and disaster recovery features depend on the capabilities of each Service. Depending on what is supported, workloads can be deployed in active/active or active/passive mode. Load balancing and failover policies can also be applied to extend resiliency beyond the boundaries of a single site.

Disaster recovery for workloads includes backup and restoration features when supported. Workloads and their associated data can be recovered in another app or even another OPCP instance. Combined with IAM and quota management, workloads can be redeployed on a separate OPCP site as part of a complete disaster recovery plan.

iv. Monitoring and Self-Healing

Integrated monitoring and alerting (Grafana, Prometheus) provide real-time detection of failures. Automated recovery workflows can restart failed services or trigger infrastructure-level remediation actions.



6. Hardware specifications

All hardware specifications are subject to evolve

a. Half-Rack

Height	24U <i>(usable space varies depending on deployment model to accommodate control and network equipment)</i>
Width	600 mm
Depth	1000 mm
Maximum load capacity	500 kg
Power capacity	5 to 12 kW (excluding GPU)
PDU	2 × Three-phase 32A – managed function + 2 × Single-phase 16A – smart function

b. Full-Rack

Height	42U to 48U <i>(usable space varies depending on deployment model to accommodate control and network equipment)</i>
Width	600 mm
Depth	1200 mm
Empty weight	140kg
Maximum load capacity	750 kg
Power capacity	5 to 15 kW (excluding GPU)
PDU	2 × Three-phase 32A – managed function + 2 × Single-phase 16A – smart function



c. Capacity ranges

The OPCP solution offers a modular hardware approach, with server ranges organized by Cloud service profiles, from lightweight configurations with 8 servers for pilot and edge services, to hyperscale clusters spanning over 150 racks of 42 servers, enabling full-stack Cloud Services, AI workloads, and sovereign infrastructure at scale.

The following table provides an example of the available server ranges (July 2025), categorized by service type.

July 2025

	# of sku	Chassis height	CPU	# of socket	CORES	RAM (GB)	GPU	DISK
COMPUTE Bare Metal VMware	3	1 or 2U	INTEL	1 or 2	From 24 to 64	From 384 to 1 024	N/A	6×3.84TB 12×3.84TB
	3	1 or 2U	AMD	1 or 2	From 32 to 96	From 512 to 1 536	N/A	18×3.84TB 24×3.84TB
COMPUTE OPENSTACK	2	2U	AMD	1 or 2	96	256	N/A	6×7.84TB 12×7.84TB
BLOCK STORAGE	1	2U	INTEL		48	256	N/A	18×7.84TB 24×7.84TB
OBJECT STORAGE	1	4U	INTEL		32	128	N/A	33x16
NUTANIX	6	1 or 2U	INTEL	1 or 2	From 16 to 48	From 128 to 798	N/A	6×3.84TB 12×3.84TB 18×3.84TB 24×3.84TB 6×7.84TB 12×7.84TB 18×7.84TB 24×7.84TB
SAP-HANA	3	2U	INTEL	1 or 2	From 32 to 64	From 512 to 1 024	N/A	3x1.6 + 6x3.84
GPU	2	2U	INTEL	1 or 2	From 32 to 48	From 512 to 1 024	Nvidia L40S or H200	4x1.92TB 4x3.84TB

OVHcloud presales experts support customers in defining the optimal hardware architecture based on their functional, performance, and compliance requirements.

[Learn more by exploring OPCP Core Datasheet.](#)

7. Operating model

The OPCP Operating Model defines the roles, responsibilities, and service boundaries for deploying and operating the On-Prem Cloud Platform across different customer needs.

a. Connectivity

In OPCP, the chosen management model directly impacts the level of connectivity required between the OPCP and OVHcloud.

First, the control plane is by design deployed on-premises, allowing the platform and its workloads to operate autonomously, even in environments with no external network connection. This architectural choice ensures operational continuity in all cases

Separately, the management model determines whether an external connection is required:

- **When managed by the customer**, the solution can remain completely disconnected, maintaining full local control.
- **When managed by OVHcloud**, a permanent secure connection is required to enable remote operations.

b. Operations

OPCP operations are structured around four key areas to ensure end-to-end platform reliability and support.

- Solution management
- Customer support
- Software engineering
- Hardware maintenance



c. Management model and support

Regardless of who manages the solution, OVHcloud will always provide support and software engineering - and oversee hardware maintenance

	Solution management	Customer support	Software engineering	Hardware maintenance
Customer-managed	✗	✓	✓	✓
OVHcloud Managed	✓	✓	✓	✓

 Provided by OVHcloud
 not provided by OVHcloud

For a detailed overview of OPCP model and support, please refer to the dedicated brochure.

d. Professional services

OVHcloud's Professional Services team supports customers throughout their OPCP journey by providing expert guidance and hands-on assistance at every stage of the project.

From cloud strategy and organizational consulting to IT asset mapping and analysis, they help define a tailored approach aligned with business goals. Services include architecture design, deployment, and migration support, as well as ongoing maintenance and compliance advisory. Security, innovation enablement, and custom training programs are also provided to ensure long-term success and autonomy in managing the On-Prem Cloud Platform.

For a detailed overview of OPCP Professional Services offers, please refer to the dedicated brochure.

8. Interoperability & Sovereignty

The On-Prem Cloud is designed to meet modern requirements for openness, interoperability, and digital sovereignty. It ensures seamless integration with existing OVHcloud ecosystems while preserving customer autonomy, transparency, and compliance with regulatory standards.

a. Compatible with OVHcloud public API & Standards

OPCP is natively aligned with the APIs and operational models used in OVHcloud's public and Private cloud.

Customers can use the same tools, automation scripts (Terraform, Ansible), and management interfaces across both on-prem and public deployments, enabling hybrid multi-environment operations (including SecNumCloud env.). This standardization ensures easy portability of workloads and simplifies operations for teams already trained on OVHcloud technologies.

b. No Proprietary lock-in

OPCP is fully built on open-source foundations and open standards, with no dependency on proprietary protocols, or formats.

All components, including the infrastructure orchestration layer, service catalog, and monitoring stack, can be audited, replaced, or extended by the customer.

This openness ensures long-term flexibility, freedom of evolution, and avoids vendor lock-in. Customers remain in full control of their platform and can adapt it to their specific needs, governance models, or future strategic choices, without being tied to a closed ecosystem.

c. Security & Compliance

Security and regulatory compliance are foundational principles of the On-Prem Cloud Platform, both by design and by default. The platform implements industry-recognized hardening and monitoring practices, and aligns with sovereign requirements such as ANSSI's SecNumCloud, making it suitable for critical and regulated environments.

OPCP is the same core platform architecture used internally by OVHcloud to deliver its own services—services that are already qualified or certified under major compliance frameworks such as **SecNumCloud, ISO/IEC 27001, and HDS (Health Data Hosting)**. This gives customers a strong trust anchor and concrete evidence of the platform's robustness and auditability.

Key built-in security features include:

- **Full Disk Encryption (SED):** All data disks are encrypted, with passphrases securely stored in a Key Management System (KMS).
- **Hardened Operating Systems:** Secured using CIS and ANSSI guidelines
- **Malware Detection:** Periodic file system analysis with alerting capabilities if logs are integrated with a SIEM.
- **Log Management:** Centralized collection and forwarding of logs to the customer's SIEM, ensuring traceability and audit-readiness.
- **SSO and IAM via Keycloak:** Supports role-based access control, segmentation of duty, and identity federation with the customer's existing directory services.
- **Secret Management**

This comprehensive security posture enables customers to deploy OPCP in sensitive contexts (healthcare, government, finance, defense), while benefiting from the maturity of a platform already proven in sovereign and regulated environments.

9. Getting started

To support customers in successfully adopting the On-Prem Cloud Platform, OVHcloud provides a structured onboarding framework covering all key phases of the deployment journey, from initial scoping to operational delivery. This includes technical and business assessments, architecture design, PoC execution, and full-scale deployment. Each step is guided by experts to ensure alignment with the customer's goals, constraints, and compliance requirements.

For a detailed description of the onboarding methodology, including the **Assessment, Design & Deployment Plan, Proof of Concept, and Delivery process**, please refer to the brochure: [OPCP The journey](#).

End of document